

A new methodology for efficiency calibration of uncharacterized Ge detectors by using characterized Ge detectors

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ABSTRACT

The measurement of gamma emitters natural radionuclides is a very relevant subject in environmental radioactivity in many fields. For this, a valid calibration based on the full-energy peak efficiency (*FEPE*) is essential, which can be carried out for gamma-ray spectrometry using Monte Carlo codes (Geant4, LabSOCS, MNCP, etc.) without needing a source preparation and with much less time consuming compared to experimental methodologies. Therefore, this work aims to develop a methodology for the *FEPE* determination of uncharacterized Ge detectors by using characterized Ge detectors with known *FEPE*. For this, a general relationship was found between the simulated *FEPE* (ϵ^{sim}) for characterized Ge detectors by LabSOCS and the experimental one (ϵ^{exp}) for uncharacterized Ge detectors obtained for a problem sample (phosphate rock (PR) in our case), where ϵ^{sim} and ϵ^{exp} were factorized using three correction factors: geometrical, photon self-absorption and true coincidence summing (TCS) effects. Thus, the $\epsilon^{sim}/\epsilon^{exp}$ ratio (*FEPER*) was obtained for two couples of Ge detectors (2 characterized and 1 uncharacterized detectors), using 4 different cylindrical geometries (C) and 2 Marinelli (M) ones, and varying the sample thickness (h), and gamma energy (E_γ) (from 46 keV (^{210}Pb) to 1765 keV (^{214}Bi)). The consistency obtained for the *FEPERs* for C geometries suggested a constant geometric factor, finding some deviations for M geometries only at $E_\gamma < 150$ keV. Given a pair of detectors, the behavior of the *FEPERs* was found to be very similar for all geometries, and h at each specific E_γ . Moreover, a constant relationship between the *FEPERs* of both detector couples versus E_γ was found. Then, the influence of the TCS effects was also studied varying the distance between the geometry bottom and detector window. The developed methodology was also validated confirming the consistency of the *FEPERs*, which were found to be independent on the chemical composition and apparent density of the sample, supporting our theoretical hypothesis. Consequently, given that the *FEPER* was found to be independent on the sample type and h , using our developed methodology it is possible to obtain ϵ^{exp} for any sample, geometry and h using the *FEPER*, previously obtained at each E_γ for the sample selected for methodology development (PR in our case), and the ϵ^{sim} that is obtained for the desired case.

1. Introduction

The measurement of natural radionuclides such as $^{234,228}\text{Th}$, $^{228,226}\text{Ra}$, $^{214,212,210}\text{Pb}$, $^{214,212,210}\text{Bi}$ and ^{40}K is essential for multiple fields of research in environmental radioactivity, and can be applied to radiological control, tracing and residence times of atmospheric aerosol masses, sediment dating, declassification of building materials, among many other applications (Abril, 2023; Appleby and Oldfield, 1983; Barba-Lobo, et al., 2023a; Barba-Lobo et al., 2022; Baskaran, 2011). For

this, it is necessary to choose an appropriate radiometric technique, being gamma-ray spectrometry one of the most used worldwide. The full-energy peak efficiency (*FEPE*) needs to be determined with precision and accuracy for a wide variety of sample matrix types and geometries, as well as for a wide range of energies.

In the case of the efficiency calibration to obtain the *FEPE* in the calibration matrix (ϵ_c) and, subsequently, in any problem sample (ϵ), the use of different types of software in gamma-ray spectrometry has become widespread such as Geant4 (Arce et al., 2014; Dziri et al., 2012;

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Quintana and Montes, 2014), LabSOCS (Suárez-Navarro et al., 2020), MCNP (Maurotto et al., 2009), EFFTRAN (Vidmar et al., 2011), PENELOPE (Guerra J et al., 2018; Guerra J et al., 2015; Peyres and García-Toraño, 2007) and DETEFF (Díaz and Vargas, 2010; Díaz and Vargas, 2008), are generally based on Monte Carlo code. One of the great advantages offered by the use of those software compared to experimental methodologies for the efficiency calibration is the possibility of obtaining ε_c for a wide variety of geometries and types of matrices in much less time. This is because to obtain ε_c in a simulated way, the preparation and experimental measurement of calibration standards are not necessary. In order to obtain the FEPE in a simulated way using the software mentioned above, it is necessary to carry out a prior characterization of the gamma detector.

Furthermore, using those software, it is also possible to transform ε_c to ε , for which various types of ε_c corrections are usually used due to: 1. The geometric conditions for which the efficiency calibration is being carried out, 2. the differences in photon attenuation caused by differences in compositions and apparent densities between the calibration standard and problem sample, and 3. the true coincidence summing (TCS) effects present for various gamma emissions belonging to natural radionuclides, especially those corresponding to ^{214}Bi and ^{208}Tl (Gilmore and Hemingway, 1995; Tedjani et al., 2016; Venegas-Argumedo and Montero-Cabrera, 2015).

Taking into account everything previously mentioned, the main objective of this study is to develop an efficiency calibration for uncharacterized Ge detectors by using Ge detectors characterized with analogous technology and geometry (that is, detectors of extended range (XiRa) type and coaxial in our case) for a wide variety of geometries and types of matrices, as well as for a wide range of energies, using LabSOCS in the case of the simulated FEPEs (characterized detectors). Calibrating a Ge detector for many geometries is an expensive process, both in operator time and materials. To do this, given a specific geometry and a pair of detectors (characterized and uncharacterized), we establish the hypothesis of the existence of a functional relationship between the FEPEs of both detectors. To the best of our knowledge, this is the first study to carry out this type of efficiency calibration for uncharacterized gamma detectors, where this study has a great applicability worldwide for numerous laboratories dedicated to the measurement of environmental radioactivity.

2. Methods and materials

2.1. Method

In the case of the FEPE for any problem sample (ε), the hypothesis that it can be factored can be established:

$$\varepsilon = \varepsilon_c f_g f_a f_s \quad (1)$$

where ε_c is the FEPE obtained in the calibration sample matrix, and f_g , f_a and f_s are the corrections due to geometric conditions, photon self-absorption effects and true coincidence summing (TCS) effects, respectively. Thus, in Eq. (1) it is assumed that it is possible to carry out the factorization of the different corrections of ε_c , which is a valid assumption as proven in previous works (Barba-Lobo and Bolívar, 2022).

The FEPE obtained for any problem sample using LabSOCS (characterized detectors) and through experimental methodologies (uncharacterized detectors) could be denoted as ε^{sim} and ε^{exp} , respectively. In the case of the ε^{exp} determination with the same geometry, the following equation was employed:

$$\varepsilon^{exp} = \varepsilon_c^{exp} f_a = \frac{G - B - F - I}{P_\gamma a m t} f_a \quad (2)$$

where G , B , F and I are the total number (gross) of counts for the full-energy peak of interest whose centroid is E_γ , Compton continuum,

background and interference term, respectively, a_c and m_c are the activity concentration of each radionuclide and the mass in the case of the calibration standard, respectively, t is the counting time and P_γ is the gamma emission probability of each E_γ , which were taken from DDEP, 2017.

In the case of the f_a factor used in Eq. (2), the Cutshall model was applied in our case (Cutshall et al., 1983). In Eq. (2), $f_g = 1$ since ε_c^{exp} and ε^{exp} were obtained for the same detector (DET-3 in our case) and under the same geometrical conditions. In addition, it is important to clarify that the calculation of ε^{exp} was carried out without including corrections due to TCS effects (f_s) since in the case of the ε^{sim} , they were not included, where $f_s = 1$ for energies not affected by TCS effects.

In the case of the ε^{sim} calculations, the Geometry Composer software (version 4.4.1) from Mirion Industries was used (Canberra Industries, 2012), with the Beaker Editor option being employed to define the outline of each geometry used in this study. From this measurement geometry, the FEPE was calculated as a function of energy, taking into account both the exact dimensions of the cylindrical containers and the chemical compositions of these containers, together with the selected experimental matrices.

The methodologies used to obtain the FEPEs through LabSOCS as well as through experimental methodologies have been developed by Barba-Lobo et al. (2023b) and Barba-Lobo et al. (2021a), respectively.

Taking into account Eq. (1), the ratio of FEPEs (FEPER), $\varepsilon^{sim} / \varepsilon^{exp}$, for a specific energy, E_γ , could be written as follows:

$$FEPER = \frac{\varepsilon^{sim}}{\varepsilon^{exp}} = \frac{\varepsilon_c^{sim} f_g^{sim} f_a^{sim} f_s^{sim}}{\varepsilon_c^{exp} f_g^{exp} f_a^{exp} f_s^{exp}} = \frac{\varepsilon_c^{sim} C_g f_s^{sim}}{\varepsilon_c^{exp} f_s^{exp}} \quad (3)$$

where it has been assumed that $f_g^{sim} / f_g^{exp} = cte = C_g$ for each geometry and $f_a^{sim} = f_a^{exp}$. This is possible if the measurements with the characterized and uncharacterized detectors are made under the same geometric conditions (same distance between geometry bottom and detector window). The second assumption is possible if the same sample is used, that is, the same apparent density (ρ), thickness (h) and chemical composition. Moreover, note that, in the case of corrections due to TCS effects, it is not possible to assume equality between f_s^{sim} and f_s^{exp} , since this correction type depends on the gamma detector used.

In the case of energies that are not affected by TCS effects, it is possible to rewrite Eq. (3) as follows:

$$FEPER = C_g \frac{\varepsilon_c^{sim}}{\varepsilon_c^{exp}} \quad (4)$$

Taking into account Eq. (3) for the same energy, $\varepsilon^{sim} / \varepsilon^{exp}$ would be independent of the problem sample selected for the measurement, as well as its thickness (h) for the same geometry. Consequently, it is necessary to find the existing $\varepsilon^{sim} / \varepsilon^{exp}$ relationship considering a great diversity of sample matrix types and geometries.

Consequently, by Eqs. (3) and (4) it is possible to observe that the FEPER obtention for any problem sample with any thickness (h) will not require any application of experimental self-absorption correction (f_a^{exp}), which is one of the several advantages offered by the methodology developed in this study.

In the case of Eqs. (3) and (4), they have been shown in order to theoretically prove the initial hypothesis, which is experimentally demonstrate in Section 3. Thus, in the case of the ε^{exp} calculation, it is possible to be carried out directly by the following simpler version of Eq. (2):

$$\varepsilon^{exp} = \frac{G - B - F - I}{P_\gamma a m t} \quad (5)$$

where a and m are the activity concentration of each radionuclide and the mass, respectively, in the case of the problem sample.

2.2. Materials

Regarding the type of sample used in this study, a phosphate rock sample (code PR) was selected, composed mainly of fluorapatite ($\text{Ca}_5(\text{PO}_4)_3\text{F}$) and with ρ around 1.6 g cm^{-3} , whose activity concentration of the radionuclides belonging to the ^{238}U series is $1569(26) \text{ Bq kg}^{-1}$, with all radionuclides of the ^{238}U series in secular equilibrium, as expected for this type of sample (Pérez-Moreno et al., 2002).

In addition, with regard to the validation of the methodology developed in this study, a certified reference material (CRM) (code RGU-1) was used, provided by the IAEA (International Atomic Energy Agency), with a majority composition of SiO_2 , with ρ about 1.6 g cm^{-3} and a reference activity concentration of $4940(15) \text{ Bq kg}^{-1}$ for radionuclides belonging to the ^{238}U series (IAEA, 1987; IAEA, 2020).

Moreover, in order to carry out the validation of the developed methodology, a non-certified sample (code Sn) was also employed which is a tin sample mainly formed by Sn (53%), O (24%), W (7%), Zr (5%), Fe (3%) and Co (3%), whose apparent density is about 3.5 g cm^{-3} and reference activity concentration for the ^{238}U -series radionuclides is $4989(58) \text{ Bq kg}^{-1}$ (Barba-Lobo and Bolívar, 2022).

For the methodology validation, another sample was employed, a phosphogypsum (code PG), which was employed for a proficiency test (CSN-CIEMAT, 2008), whose reference concentrations of ^{238}U , ^{226}Ra and ^{210}Pb (in Bq kg^{-1}) are $48.6(1.2)$, $634(13)$ and $781(15)$, respectively. The chemical composition of PG is mainly formed by H (2%), O (55%), S (18%), Ca (22%), Si (1%), Al (1%), P (0.5%) and F (0.5%), with ρ around 1.3 g cm^{-3} . CSN and CIEMAT are the acronyms for the Spanish Nuclear Safety Council (“Consejo de Seguridad Nuclear”) and the Centre for Energy, Environment and Technology Research (“Centro de Investigaciones Energéticas, Medioambientales y Tecnológicas”), respectively.

The different types of geometries used were cylindrical (C) and Marinelli (M) type. See Appendix A (Supplementary Material) for detailed information on the appearance and dimensions of each geometry, as well as the thicknesses used for each geometry. The code for each of them, as well as their respective external diameters (D) and heights (H), are specified below:

Cylindrical type

C72: $D = 71.8 \text{ mm}$ and $H = 83.9 \text{ mm}$; **C50:** $D = 50.3 \text{ mm}$ and $H = 10.0 \text{ mm}$; **C36:** $D = 36.1 \text{ mm}$ and $H = 65.1 \text{ mm}$; **C29:** $D = 29.0 \text{ mm}$ and $H = 69.9 \text{ mm}$.

Marinelli type

M128: $D = 127.6 \text{ mm}$ and $H = 129.1 \text{ mm}$; **M121:** $D = 120.7 \text{ mm}$ and $H = 118.1 \text{ mm}$.

For both the cylindrical type (C) and the Marinelli type (M) geometries, $\epsilon^{\text{sim}}/\epsilon^{\text{exp}}$ was obtained for the PR sample, varying the thickness of the sample (h) and fixing the energy (E_γ) at the following values corresponding to ^{238}U series radionuclides: 46 keV (^{210}Pb), 63 keV (^{234}Th), 186 keV ($^{226}\text{Ra} + ^{235}\text{U}$), 295 keV (^{214}Pb), 352 keV (^{214}Pb), 609 keV (^{214}Bi), 1120 keV (^{214}Bi) and 1765 keV (^{214}Bi), thus being able to cover a wide range of thicknesses and energies for each geometry.

2.3. Gamma detectors

Regarding the gamma detectors used in this study, high purity germanium (HPGe) detectors are selected for both characterized and uncharacterized detectors provided by Mirion Industries. Three extended range (XtRa) detectors were employed, where the characterized ones have the codes DET-1 and DET-2, and DET-3 for the uncharacterized one, whose relative efficiencies with respect to a $3'' \times 3''$ NaI(Tl) detector at 1332 keV are 42.1% and 115.7% , and 38.4% , respectively. Regarding the electronic chain connected to each detector, these were of conventional type, using the Genie 2000 software (Canberra Industries,

2004) for data acquisition and spectrum analysis for the three detectors. DET-3 is not characterized, but ϵ_c^{exp} is known.

The methodology was developed using the different geometries and energies previously mentioned in Section 2.2, selecting the PR sample in this case. Thus, the calculations of the FEPE ratios ($FEPER$), $\epsilon^{\text{sim}}/\epsilon^{\text{exp}}$, were carried out for both couples of detectors, that is, DET-1 and DET-3, and DET-2 and DET-3, obtaining $FEPER_1$ and $FEPER_2$, respectively.

3. Results and discussion

3.1. Obtention of the FEPE ratios

The $FEPER$ s were calculated using Eq. (3) for the geometries C72, C50, C36, C29, M128 and M121 in the case of the PR reference sample, varying the thickness (h) and energy (E_γ). Thus, 5 h values were selected for each geometry, which are well distributed throughout each geometry, excepting for C50 geometry, for which only 1 h value was selected due to its relatively low height. In the case of the M128 and M121 geometries, a h value was selected just over the detector window and another h value under the detector window (see Appendix A in Supplementary Material) in order to test the methodology validity for extreme geometric cases.

As can be seen in Fig. 1 (on the left), the $FEPER_1$ values obtained varying h for C29 geometry were very similar from each other for each E_γ in the case of the couple of detectors DET-1 – DET-3. The similar behavior of $FEPER_1$ for all the h values for each E_γ can be explained by using Eq. (4). The average $FEPER_1$ value were also shown for each energy in order to make easier the visualization of the similarity previously mentioned. Thus, by using Eq. (4), the ratio of $FEPER$ s for a specific energy and for two different thicknesses (h_1 and h_2) would be written as $\frac{\epsilon_c^{\text{sim}}(h_2)}{\epsilon_c^{\text{exp}}(h_2)} \cdot \frac{\epsilon_c^{\text{exp}}(h_1)}{\epsilon_c^{\text{sim}}(h_1)}$, that is, as the ratio of $FEPER$ s obtained in the calibration matrix. Therefore, the ϵ_c^{exp} and ϵ_c^{sim} curves versus h must have a similar behavior making that the relative difference $\epsilon_c^{\text{exp}}(h)$ and $\epsilon_c^{\text{sim}}(h)$ be the same for each h value. This explains why the behavior of $FEPER_1$ is the same at each specific E_γ regardless of the h value selected.

In addition, the $FEPER_1$ values varying E_γ are very similar from each other which is consistent due to DET-1 and DET-3 have very similar relative efficiencies (42.1% and 38.4% , respectively). Note that for 609 keV , 1120 keV and 1765 keV the $FEPER_1$ values are some different from those obtained for the other E_γ , which is due to the presence of true coincidence summing (TCS) effects for these three energies. In addition, note that for low energies (46 keV and 63 keV), the $FEPER_1$ values are less than 1, while relative efficiency for DET-1 is higher than DET-3 one. This can be explained due to the self-absorption that occurs in the Ge detector itself, where the self-absorption in the Ge detector itself for the DET-1 must be higher than for the DET-3, causing the higher self-absorption for low energies (Barba-Lobo et al., 2021b; Bonczyk, 2018; Długosz-Lisiecka and Ziomek, 2015) and, consequently, $FEPER_1 < 1$ for 46 keV and 63 keV .

For the other couple of detectors, that is, DET-2 – DET-3, in the case of geometry C29 (Fig. 1 on the right), the $FEPER_2$ have also similar values for each specific E_γ regardless of the h value selected, as previously occurred for $FEPER_1$. However, the $FEPER_2$ values obtained varying E_γ are very different from each other, unlike that occurs for $FEPER_1$. This is consistent since the relative efficiencies of DET-2 and DET-3 are very different from each other (115.7% and 38.4% , respectively), causing this dependence of the $FEPER_2$ on E_γ .

Regarding the geometries C36, C50 and C72 (Figs. 2–4, respectively), the behavior of the $FEPER_1$ and $FEPER_2$ versus E_γ for each h value is very similar to that previously found and discussed for geometry C29. The obtained $FEPER_1$ and $FEPER_2$ values were very similar from each other for all the geometries, where $FEPER_1$ and $FEPER_2$ were ranged from about 0.9 (46 keV and 63 keV) to 1.2 (609 keV and 1120 keV), and from about 1.1 (46 keV and 63 keV) to 2.8 (609 keV and 1120 keV), respectively. Only for $FEPER_2$ in the case of C50 geometry (Fig. 3 on the

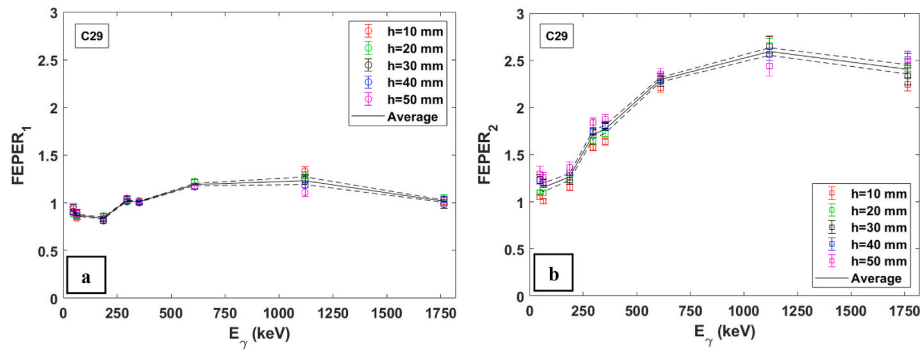


Fig. 1. FEPE ratios ($FEPEr$) obtained at specific thicknesses (h) varying the energy (E_γ), using cylindrical geometry C29. The $FEPEr$ ($FEPEr_1$ (a) and $FEPEr_2$ (b)) were obtained comparing the $FEPEs$ simulated by LabSOCS for characterized detectors (DET-1 and DET-2, respectively) to the experimental $FEPEs$ obtained for a non-characterized detector (DET-3). In the case of the average $FEPEr$ (solid line), its uncertainty at 1 sigma level was also shown (dashed line).

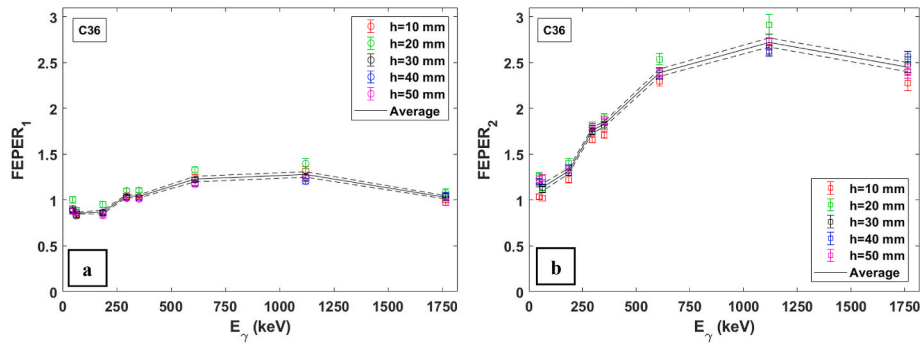


Fig. 2. FEPE ratios ($FEPEr$) obtained at specific thicknesses (h) varying the energy (E_γ), using cylindrical geometry C36. The $FEPEr$ ($FEPEr_1$ (a) and $FEPEr_2$ (b)) were obtained comparing the $FEPEs$ simulated by LabSOCS for characterized detectors (DET-1 and DET-2, respectively) to the experimental $FEPEs$ obtained for a non-characterized detector (DET-3). In the case of the average $FEPEr$ (solid line), its uncertainty at 1 sigma level was also shown (dashed line).

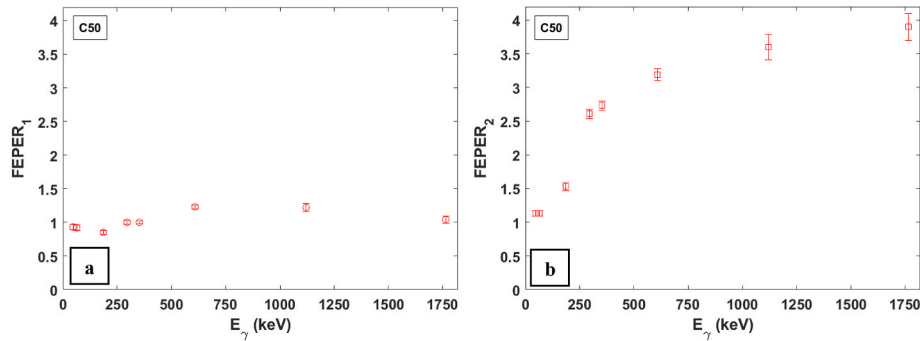


Fig. 3. FEPE ratios ($FEPEr$) obtained at specific thicknesses ($h = 7.29$ mm) varying the energy (E_γ), using cylindrical geometry C50. The $FEPEr$ ($FEPEr_1$ (a) and $FEPEr_2$ (b)) were obtained comparing the $FEPEs$ simulated by LabSOCS for characterized detectors (DET-1 and DET-2, respectively) to the experimental $FEPEs$ obtained for a non-characterized detector (DET-3).

right), the $FEPEr_2$ range was somehow different from that found for the other 3 geometries, reaching a maximum $FEPEr_2$ value about 4 for C50 geometry. The fact that the obtained $FEPEr_1$ and $FEPEr_2$ values were very similar in general for the cylindrical geometries means that the constant C_g related to the geometrical factor that was shown in Eq. (4) must have a similar value for these geometries regardless of the selected h and E_γ .

Then, with respect to the geometries of Marinelli type, that is, M121 and M128, the $FEPEr_1$ and $FEPEr_2$ are shown in Figs. 5 and 6, respectively. Note that in the case of the M121, the $FEPEr_2$ was not calculated due to the M121 geometry was not possible to be used for DET-2 because of incompatibilities between their dimensions.

For geometry M121, the $FEPEr_1$ behavior was similar to that found

for the cylindrical geometries, excepting for the lowest thickness ($h = 35.45$ mm), for which it is possible to observe that the $FEPEr_1$ values deviated from the average when varying the energy. This is consistent since for this extreme geometrical case, h is so low that it is very complicated that the gamma photons reach the detector in the same way than for the other thicknesses. Consequently, the $FEPEr_1$ values deviated from the average for $h = 35.45$ mm, where the $FEPEr_1$ values were lower the average for low energies (46 keV and 63 keV), while for high energies (> 150 keV) the $FEPEr_1$ values were upper the average. This agrees well with the discussion previously provided for the cylindrical geometries in relation to the higher self-absorption for 46 keV and 63 keV in the case of the DET-1 window compared to the DET-3.

Regarding the geometry M128, the $FEPEr_1$ followed the same

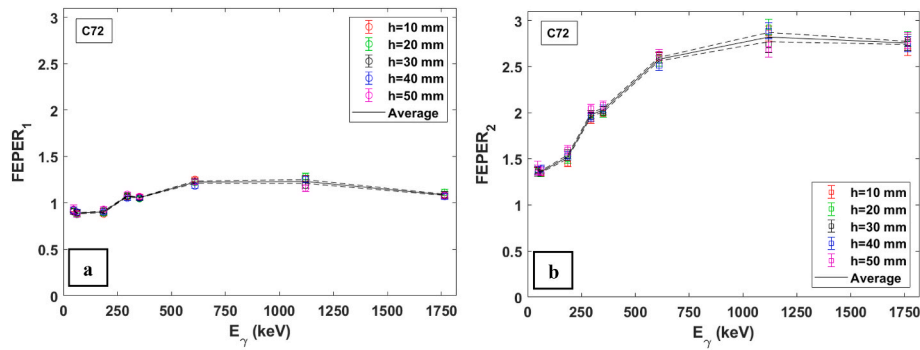


Fig. 4. FEPE ratios ($FEPE$) obtained at specific thicknesses (h) varying the energy (E_γ), using cylindrical geometry C72. The $FEPE$ ($FEPE_1$ (a) and $FEPE_2$ (b)) were obtained comparing the $FEPE$ s simulated by LabSOCS for characterized detectors (DET-1 and DET-2, respectively) to the experimental $FEPE$ s obtained for a non-characterized detector (DET-3). In the case of the average $FEPE$ (solid line), its uncertainty at 1 sigma level was also shown (dashed line).

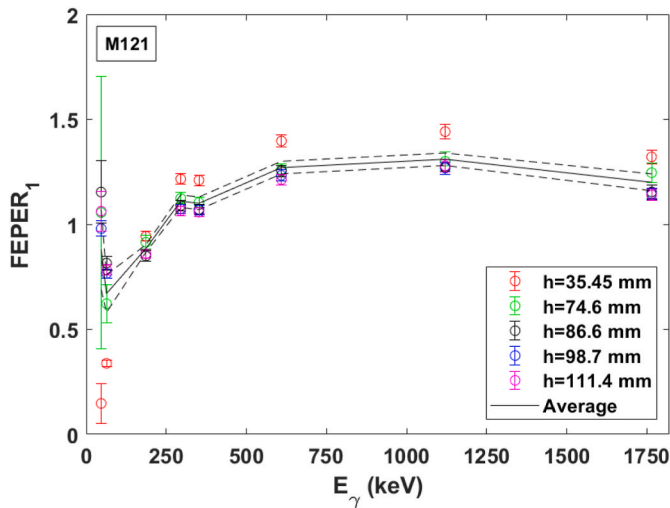


Fig. 5. FEPE ratios ($FEPE$) obtained at specific thicknesses (h) varying the energy (E_γ), using Marinelli geometry M121. The $FEPE$ ($FEPE_1$) were obtained comparing the $FEPE$ s simulated by LabSOCS for characterized detectors (DET-1) to the experimental $FEPE$ s obtained for a non-characterized detector (DET-3). In the case of the average $FEPE$ (solid line), its uncertainty at 1 sigma level was also shown (dashed line).

behavior than that obtained for geometry M121, where for the lowest thickness ($h = 41.28$ mm in this case), the $FEPE_1$ values deviated again from the average in the same way than that occurred for M121, that is, the $FEPE_1$ values were lower and upper the average for low and high energies, respectively. This agrees well with the explanation previously

carried out for geometry M121. Furthermore, the $FEPE_1$ values were similar to those obtained for geometry M121 as well as for the cylindrical geometries, ranging from about 0.8 (46 keV and 63 keV) to 1.2 (609 keV and 1120 keV) for geometry M128. Then, for the $FEPE_2$ values obtained for geometry M128, they were also out of the average for $h = 41.28$ mm. In addition, the $FEPE_2$ behavior found for geometry M128 for the other h values was very similar to that observed for the $FEPE_2$ in the case of the cylindrical geometries.

3.2. Obtention of a general FEPE for each energy

Due to the similarity found among the $FEPE$ s previously obtained in Section 3.1 varying the geometry and thickness, this Section is focused on finding a general $FEPE$ for each energy that can be used for any thickness and geometry.

Firstly, in order to prove the similarity among the $FEPE$ s obtained for the different cylindrical and Marinelli geometries, the average $FEPE$ s values obtained over all the thicknesses for each geometry, $\langle FEPE \rangle$, were divided between the average $FEPE$ s obtained in the case of the geometry C36, $\langle FEPE^{C36} \rangle$. The geometry C36 was fixed in this case in order to make the comparison, obtaining $\langle FEPE_1^i \rangle / \langle FEPE_1^{C36} \rangle$ and $\langle FEPE_2^i \rangle / \langle FEPE_2^{C36} \rangle$ for each couple of detectors, respectively (see Fig. 7). As can be seen in Fig. 7, the $\langle FEPE_1^i \rangle / \langle FEPE_1^{C36} \rangle$ values were statistically compatible with 1 for all the energies. Then, in the case of the $\langle FEPE_2^i \rangle / \langle FEPE_2^{C36} \rangle$, the geometries C29, and C72 were more similar to the geometry C36 for all the energies, finding some differences between geometries C50 and C36, occurring the same between geometries M128 and C36. In addition, it is also possible to observe that the $\langle FEPE_2^{M128} \rangle / \langle FEPE_2^{C36} \rangle$ and $\langle FEPE_2^{M128} \rangle / \langle FEPE_2^{C36} \rangle$ values were very similar form each other for high energies, which agrees well with the $FEPE_2$ ranges previously

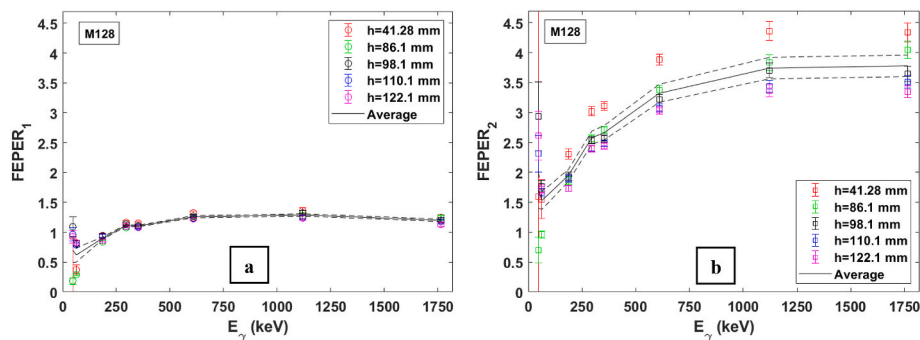


Fig. 6. FEPE ratios ($FEPE$) obtained at specific thicknesses (h) varying the energy (E_γ), using Marinelli geometry M128. The $FEPE$ ($FEPE_1$ (a) and $FEPE_2$ (b)) were obtained comparing the $FEPE$ s simulated by LabSOCS for characterized detectors (DET-1 and DET-2, respectively) to the experimental $FEPE$ s obtained for a non-characterized detector (DET-3). In the case of the average $FEPE$ (solid line), its uncertainty at 1 sigma level was also shown (dashed line).

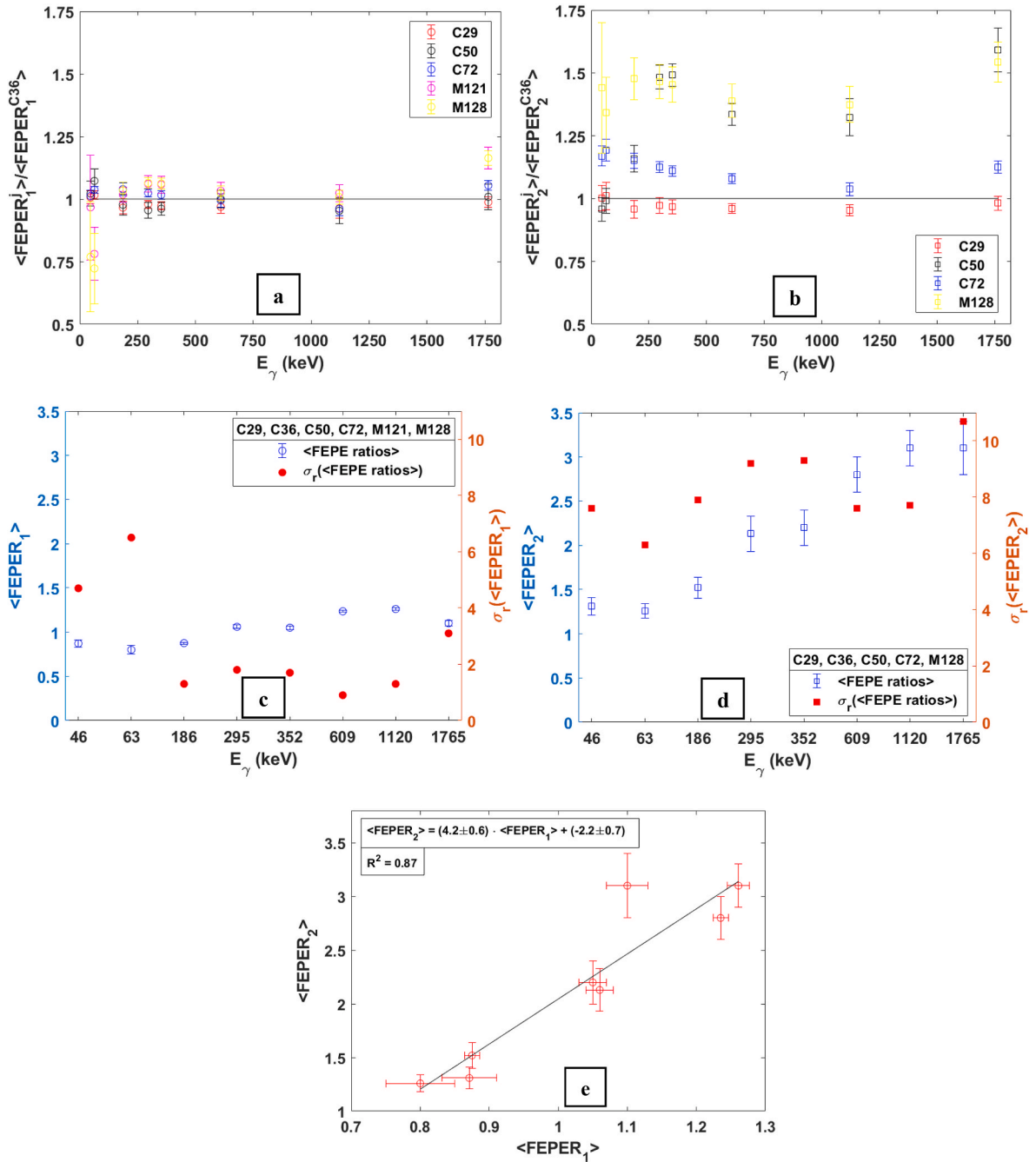


Fig. 7. Comparison between the average FEPE ratios ($\langle FEPE_1 \rangle$) (a), ($\langle FEPE_2 \rangle$) (b), obtained for each geometry (j) and the geometry C36 ($\langle FEPE_1^{C36} \rangle$, $\langle FEPE_2^{C36} \rangle$), varying the energy (E_γ), where the average FEPE ratios were calculated over all the thicknesses (h) used for each geometry. In addition, the average FEPE ratios were also obtained over all the geometries used for each couple of detectors ($\langle FEPE_1 \rangle$) (c) and ($\langle FEPE_2 \rangle$) (d), respectively) at each specific E_γ , as well as their respective relative uncertainties ($\sigma_r(\langle FEPE_1 \rangle)$ and $\sigma_r(\langle FEPE_2 \rangle)$), respectively), using an alphanumeric scale for the E_γ in this case. Moreover, $\langle FEPE_2 \rangle$ versus $\langle FEPE_1 \rangle$ and their corresponding linear fit (e) were also shown.

obtained in Section 3.1 for geometries C50 and M128.

Then, after obtaining the average FEPEs over all the thicknesses for each geometry, the FEPEs were obtained, in turn, averaging over all the geometries for each specific energy achieving $\langle FEPE_1 \rangle$ and $\langle FEPE_2 \rangle$ for each couple of detectors, respectively (see Fig. 7). Furthermore, in order to observe the deviation from the average FEPE obtained for each energy, the relative standard deviation of the average (σ_r (%)) was also plotted for each energy. As can be seen in Fig. 7, the σ_r values achieved for each energy were less than 7% for $\langle FEPE_1 \rangle$, which were between 1% and 3% for the great majority of the energies. In the case of the σ_r values obtained for $\langle FEPE_2 \rangle$, they were ranged from 6%

to 10%. Consequently, considering the σ_r values obtained for $\langle FEPE_1 \rangle$ and $\langle FEPE_2 \rangle$, it is possible to conclude that there is significant dependence of the FEPEs on the geometry type.

Furthermore, a relationship was also found between $\langle FEPE_1 \rangle$ and $\langle FEPE_2 \rangle$ as can also be seen in Fig. 7. Thus, this relationship was of linear type proving a constant behavior of $\langle FEPE_2 \rangle / \langle FEPE_1 \rangle$ versus the energy, as well as making it possible to find a general FEPE₂ function depending on FEPE₁.

As can be seen in Fig. 7, there is certain dependence of the FEPE on the energy due to the E_γ values that are affected by TCS effects, that is, 609 keV, 1120 keV and 1765 keV, where the ϵ^{exp} and ϵ^{sim} obtained

experimentally and by LabSOCS, respectively, are not corrected by TCS effects. However, it is possible to correct the ε^{exp} and ε^{sim} by TCS effects at the same time by using the methodology developed by Barba-Lobo and Bolívar (2022). For this, the distance between the geometry bottom and the detector window (d) was varied, selecting in this case d values ranged from 25 mm to 65 mm. For the TCS corrections applying this methodology, it was employed the CRM with code RGU-1 for a fixed h value of 25 mm and for geometry C36 since the ε^{exp} values obtained by Barba-Lobo and Bolívar (2022) were calculated using these conditions. Thus, the ε^{sim} were also calculated by LabSOCS for the same d and h values in the case of the RGU-1 using the detectors DET-1 and DET-2.

As can be seen in Fig. 8, the $FEPER_1(RGU-1)$ and $FEPER_2(RGU-1)$ were plotted versus d for each specific energy previously selected in Sections 3.1 and 3.2. Thus, the average $FEPER_1(RGU-1)$ and $FEPER_2(RGU-1)$ obtained over all the d values were also plotted in Fig. 8 ($\langle FEPER_1(RGU-1) \rangle$ and $\langle FEPER_2(RGU-1) \rangle$, respectively), and they were shown together with the average $FEPER$ values obtained, in turn, over all the energies (solid lines in Fig. 8). Thus, it is possible to observe that $\langle FEPER_1(RGU-1) \rangle$ values were statistically compatible with the average value obtained over all the energies, excepting in the case of 186 keV. However, since 186 keV is an energy not affected by TCS effects, it is not relevant that $\langle FEPER_1(RGU-1) \rangle$ was not statistically compatible with the average. Then, in the case of the $\langle FEPER_2(RGU-1) \rangle$, it was possible to note that after applying the methodology developed by Barba-Lobo and Bolívar (2022) to correct TCS effects, the $\langle FEPER_2(RGU-1) \rangle$ continues having a dependence on E_γ . Thus, in the

case of the $FEPER_2$ the dependence on E_γ is related to the very significant differences between the relative efficiencies of detectors DET-2 and DET-3, as previously explained in Section 3.1.

3.3. Validation of the methodology developed for the FEPERs

In Sections 3.1 and 3.2, for the methodology developed for the FEPERs, a phosphate rock (PR) sample was selected. Thus, in this Section a CRM, non-CRM and a sample used for a proficiency test, whose codes are RGU-1, Sn and PG, respectively, were chosen to test this methodology, where the Sn sample was selected due to the very high difference of its chemical composition and apparent from those related to the RGU-1 standard, allowing us to properly validate the developed methodology. For this, the geometries C36 and M128 were selected, because of the great differences between their characteristics, for the RGU-1 and Sn, and PG, respectively, according to the amount of sample available for each case. The FEPERs for the RGU-1, Sn and PG samples were calculated for four, five and three thicknesses, respectively ($h = 5$ mm, 15 mm, 20 mm and 40 mm (for RGU-1), $h = 3$ mm, 6 mm, 12 mm, 24 mm and 48 mm (for Sn sample) and $h = 41.25$ mm, 86.1 mm and 122.1 mm (for PG)), and then an average FEPER over all the h values was found for each energy using both couples of detectors, obtaining $\langle FEPER_1^{C36}(RGU-1) \rangle$ and $\langle FEPER_2^{C36}(RGU-1) \rangle$, $\langle FEPER_1^{C36}(Sn) \rangle$ and $\langle FEPER_2^{C36}(Sn) \rangle$, and $\langle FEPER_1^{M128}(PG) \rangle$ and $\langle FEPER_2^{M128}(PG) \rangle$, respectively.

The $\langle FEPER_1^{C36}(i) \rangle$ and $\langle FEPER_2^{C36}(i) \rangle$ were divided between $\langle FEPER_1^{C36}(RGU-1) \rangle$ and $\langle FEPER_2^{C36}(RGU-1) \rangle$, respectively, for each

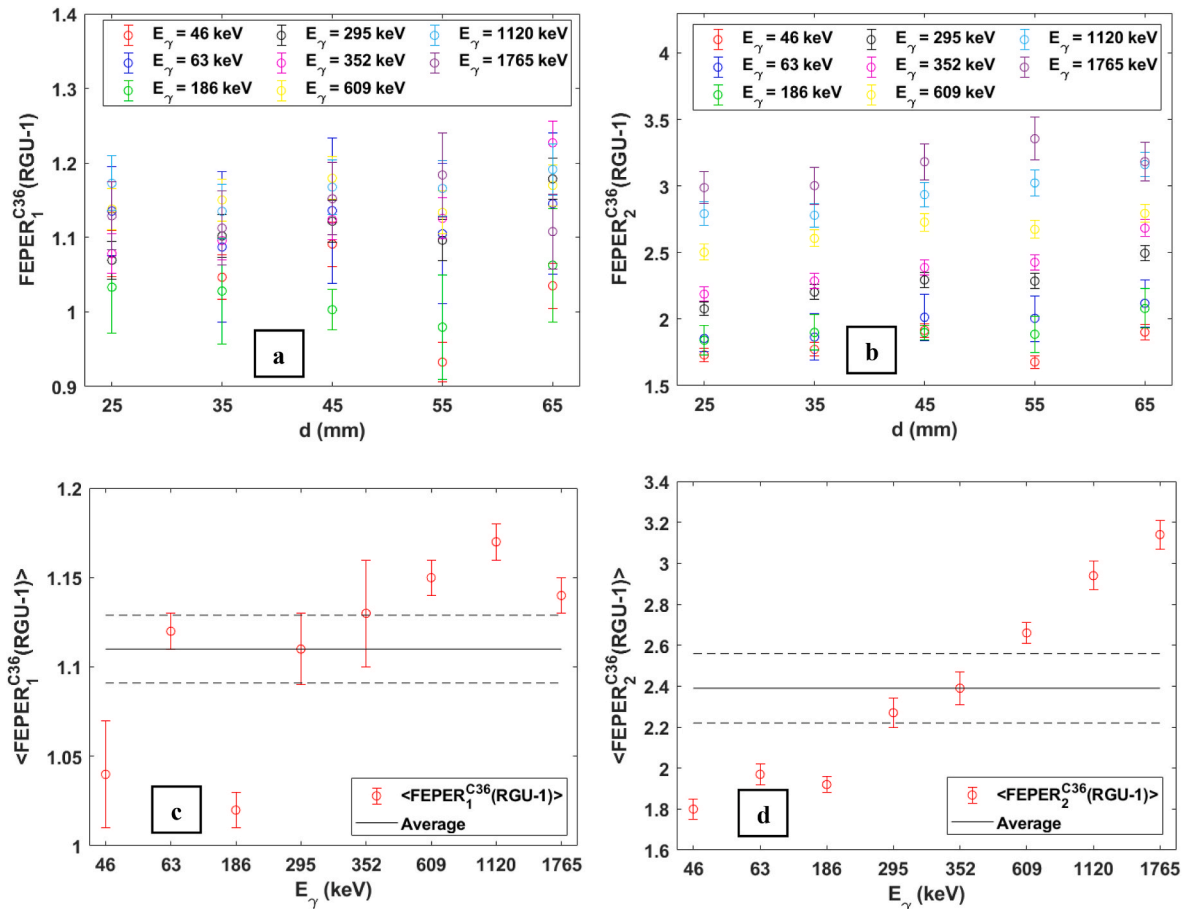


Fig. 8. Obtention of the FEPER values varying the distance between the detector window and the sample bottom (d) for each specific energy (E_γ), using the RGU-1 compacted at a fixed thickness ($h = 25$ mm) in the case of the geometry C36. These FEPER calculations were carried out for both couples of detectors, that is, the characterized detectors (DET-1 and DET-2) with the non-characterized detector (DET-3), obtaining $FEPER_1^{C36}(RGU-1)$ (a) and $FEPER_2^{C36}(RGU-1)$ (b), respectively. Then, the average FEPER values ($\langle FEPER_1^{C36}(RGU-1) \rangle$) (c) and $\langle FEPER_2^{C36}(RGU-1) \rangle$ (d)) were obtained over all the d values, which were plotted together the average FEPER value obtained over all the E_γ values.

energy selected, where $\langle FEPER_1^{C36}(i) \rangle$ and $\langle FEPER_2^{C36}(i) \rangle$ are the $FEPER_1$ and $FEPER_2$ obtained for the geometry C36 in the case of the samples i ($i = \text{PR, Sn}$) averaging over the five thicknesses used for each sample in the case of each couple of detectors (see Sections 3.1 and 3.2 for the thicknesses employed for the PR sample and Section 3.3 in the case of the Sn sample). In the case of the $\langle FEPER_1^{M128}(\text{PG}) \rangle$ and $\langle FEPER_2^{M128}(\text{PG}) \rangle$, they were divided between $\langle FEPER_1^{M128}(\text{PR}) \rangle$ and $\langle FEPER_2^{M128}(\text{PR}) \rangle$, respectively, because the PR and PG were the two samples for which was possible to fill the M128 according to the amount of sample available.

As can be seen in Fig. 9, the $\langle FEPER_1^{C36}(i) \rangle / \langle FEPER_1^{C36}(\text{RGU-1}) \rangle$ and $\langle FEPER_2^{C36}(i) \rangle / \langle FEPER_2^{C36}(\text{RGU-1}) \rangle$ as well as $\langle FEPER_1^{M128}(\text{PG}) \rangle / \langle FEPER_1^{M128}(\text{PR}) \rangle$ and $\langle FEPER_2^{M128}(\text{PG}) \rangle / \langle FEPER_2^{M128}(\text{PR}) \rangle$ were all of them statistically compatible with 1 for both samples ($i = \text{PR, Sn}$). This is consistent according to the hypothesis established in Section 2.1, where it was theoretically proven that when calculating the $FEPER$ (Eqs. (1), (3) and (4)), it was not dependent on the chemical composition or apparent density of the sample. Consequently, the $FEPER$ must be the same regardless of the selected sample since the self-absorption corrections are not needed. In addition, note that the $\langle FEPER_1^{C36}(i) \rangle$ and $\langle FEPER_1^{C36}(\text{RGU-1}) \rangle$ as well as $\langle FEPER_1^{M128}(\text{PG}) \rangle$ and $\langle FEPER_1^{M128}(\text{PR}) \rangle$ were calculated using different h values, as previously mentioned, therefore, the non-dependence of the $FEPER$ on h is also proven again.

In order to further validate the methodology, the concentrations of radionuclides were calculated for the RGU-1, Sn and PG samples using the average $FEPER$ s obtained for PR sample (Section 3.2). Thus, average $FEPER$ s obtained for each energy, and given that the ϵ^{sim} is known for each one of the three previous samples for each h , it is possible to obtain the ϵ^{exp} for each case using the relationship $FEPER = \epsilon^{sim} / \epsilon^{exp}$, where $FEPER$ is the average $FEPER$ value obtained in Section 3.2 for each energy and geometry in the case of the PR sample. Thus, knowing the ϵ^{exp} for each sample at each respective h , it was possible to obtain the concentrations of radionuclides using Eq. (5). Then, the z_{score} values were obtained comparing the calculated activity concentrations with the reference values for a proper statistical analysis of the methodology validation. As can be seen in Fig. 10, the z_{score} values were generally less than 2 for each sample, geometry and energy, using both couples of detectors (DET-1 and DET-3 on the left, and DET-2 and DET-3 on the right), which further proves the good validation of the developed methodology. This also proves again the non-dependence of the $FEPER$ on the sample type and thickness since for the RGU-1, Sn and PG samples, the concentrations of radionuclides were calculated using the $FEPER$ obtained for the PR sample. In addition, note that the z_{score} values were well distributed between the positive and negative panels demonstrating that there is no systematicity associated to the developed methodology.

4. Conclusions

This work has carried out a development of a novel and general efficiency calibration for uncharacterized gamma detectors by using characterized gamma detectors, based on the ratio of full-energy peak efficiencies ($FEPER$) of characterized detectors (ϵ^{sim} , DET-1 and DET-2) and uncharacterized detectors (ϵ^{exp} , DET-3), where ϵ^{sim} and ϵ^{exp} were factorized into three correction factors: geometrical, photon self-absorption and true coincidence effects (TCS). The developed methodology was applied to different cylindrical (C) and Marinelli (M) geometries using a rock phosphate (PR) sample, varying its thickness (h) and for a wide range of energy (E_γ). When applying the developed methodology, it is only necessary to carry out the experimental efficiency calibration of the uncharacterized Ge detector for a single sample, thickness and geometry, since the $FEPER$ was proven to be independent on sample type, thickness and geometry.

The main conclusions of this work are the following:

1. The $FEPER_1$ and $FEPER_2$ values obtained for C geometries were found to be independent on h at each E_γ , with values generally ranging from about 0.9 to 1.2, and from about 1.1 to 2.8, respectively. For M geometries, $FEPER_1$ followed similar patterns, while for $FEPER_2$ some differences were found but only at low energies ($E_\gamma < 150$ keV).
2. An insignificant dependence of the $FEPER$ s on geometry type was found, where the relative standard deviations obtained for average $FEPER$ over all the C and M geometries at each E_γ were all of them below 7% for $\langle FEPER_1 \rangle$ and between 6% and 10% for $\langle FEPER_2 \rangle$. In addition, a linear relationship was found between $\langle FEPER_1 \rangle$ and $\langle FEPER_2 \rangle$, allowing to obtain a general function for $\langle FEPER_2 \rangle$ as a function of $\langle FEPER_1 \rangle$.
3. After applying the corrections due to TCS effects varying the distance between the bottom geometry and detector window, $FEPER_1$ was observed to be independent on E_γ , while $FEPER_2$ still shew a dependence on E_γ .
4. The methodology was validated using a certified reference material (RGU-1), a non-certified reference material (Sn) and sample employed for a proficiency test (PG). The $FEPER$ s obtained using the RGU-1 were statistically compatible with those obtained in the case of the PR and Sn samples, analogously occurring between the PR and PG samples. In addition, the concentrations of radionuclides were calculated for the RGU-1, Sn and PG samples using the average $FEPER$ s obtained for the PR sample, obtaining z_{score} values generally less than 2 for each sample, geometry and energy, using both couples of detectors. This supports the theoretical hypothesis established in this study about the $FEPER$ independence on sample composition or density, confirming the no need for corrections for self-absorption. Therefore, the robustness and validity of the method has been

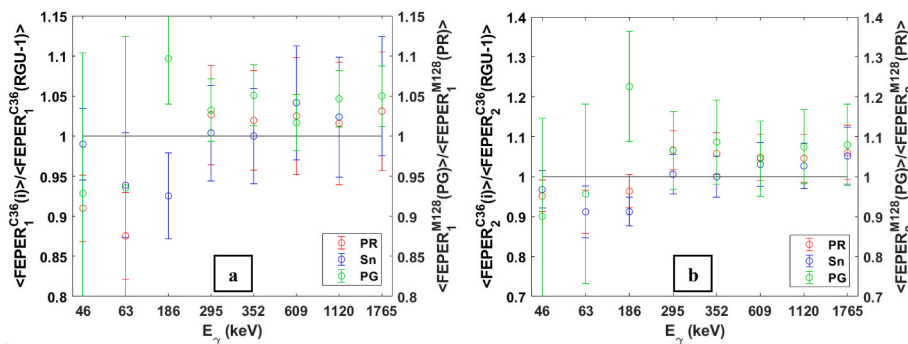


Fig. 9. Validation of the methodology developed in this study. The average $FEPER$ values obtained for the phosphate rock (PR) and the tin sample (Sn) were compared with those obtained for a certified reference material (RGU-1), where the geometry C36 was used. In the case of the sample employed for a proficiency test (PG), the comparison was carried out with respect to the phosphate rock (PR) using the geometry M128. This validation was carried out for both couples of detectors, that is, the characterized detectors (DET-1 and DET-2) with the non-characterized detector (DET-3), obtaining $\langle FEPER_1^{C36}(i) \rangle / \langle FEPER_1^{C36}(\text{RGU-1}) \rangle$ ($\langle FEPER_1^{M128}(\text{PG}) \rangle / \langle FEPER_1^{M128}(\text{PR}) \rangle$) (a) and $\langle FEPER_2^{C36}(i) \rangle / \langle FEPER_2^{C36}(\text{RGU-1}) \rangle$ ($\langle FEPER_2^{M128}(\text{PG}) \rangle / \langle FEPER_2^{M128}(\text{PR}) \rangle$) (b), respectively, where $i = \text{PR, Sn}$.

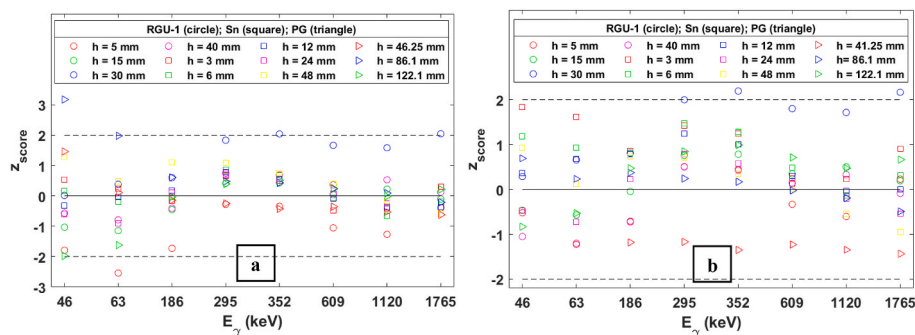


Fig. 10. Validation of the methodology developed in this study. The concentrations of radionuclides present in the RGU-1, Sn and PG samples were calculated using the average *FEPER* values obtained in the case of the PR sample for each geometry (C36 for RGU-1 and Sn, and M128 for PG). The Z_{score} values were obtained by using both couples of detectors (DET-1 and DET-3 (a), and DET-2 and DET-3 (b)).

demonstrated for a wide range of thicknesses, geometries and energies using different samples.

CRediT authorship contribution statement

A. Barba-Lobo: Writing – review & editing, Writing – original draft, Validation, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **V.M. Expósito-Suárez:** Methodology, Formal analysis, Data curation, Conceptualization. **J.A. Suárez-Navarro:** Writing – review & editing, Writing – original draft, Validation, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **J.P. Bolívar:** Writing – review & editing, Writing – original draft, Validation, Methodology, Investigation, Formal analysis, Data curation, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.radphyschem.2024.111763>.

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